



MEETING 1

Introduction

About The Company

“Healthcare Medical Group.”



Healthcare Medical Group is a private healthcare organization in Perth, Western Australia. The hospital manages thousands of patient billing cases each year through its electronic medical billing system. Recently, HMG reported growing issues with:

- Long billing turnaround times,
- Frequent rework and coding corrections, and
- Escalating administrative costs.

To address these operational challenges, HMG engaged a team of **process-mining consultants** to analyse their billing workflow using real event-log data from the hospital's ERP system.

MEETING I

Healthcare Medical Group Engaged process-mining consultants to identify operational inefficiencies, financial impacts and areas of improvements within the billing workflow

Core Business Questions



Where do Bottlenecks occur within billing flow?



How does rework inflate cost & cycle time?



Which roles are over/under-utilized?

MEETING I



The project's primary objective was to:

Use process mining to identify operational inefficiencies, quantify their financial impact, and recommend data-driven improvements to streamline the billing process.

Key goals:

- Discover the **actual billing process** as executed (not as designed).
- Identify **bottlenecks, rework loops, and SLA compliance issues**.
- Evaluate **resource utilisation** and **cost drivers**.
- Provide **targeted improvement recommendations** to reduce cycle time and costs.

What process mining does?

- Extract the data to identify real workflow
- Transform the event log by filter out irrelevant events, check, and resolve data quality issues to ensure reliable analysis
- Load the data into process mining tool (Apromore) to perform analysis
- Discover process map to visualise actual flow
- Analyse the process to identify inefficiencies
- Provide recommendations to improve the process

MEETING I

Project implementation process

1. Problem framing
2. Data collection
3. Process Analysis
4. Interpretation
5. Business impact creation

MEETING I

What we need

- Event log (xes or csv file) with timestamps
- SLA or target of the company

What's Next

After Meeting 1:

- Receive exported records from the system in standard format (CSV)
- Load the data into the process mining tool (Apromore)
- Perform analysis and summarise findings

Meeting 2:

- Present findings, discussions, and actionable recommendations



MEETING 2

Insights & Recommendations

MEETING I Recap

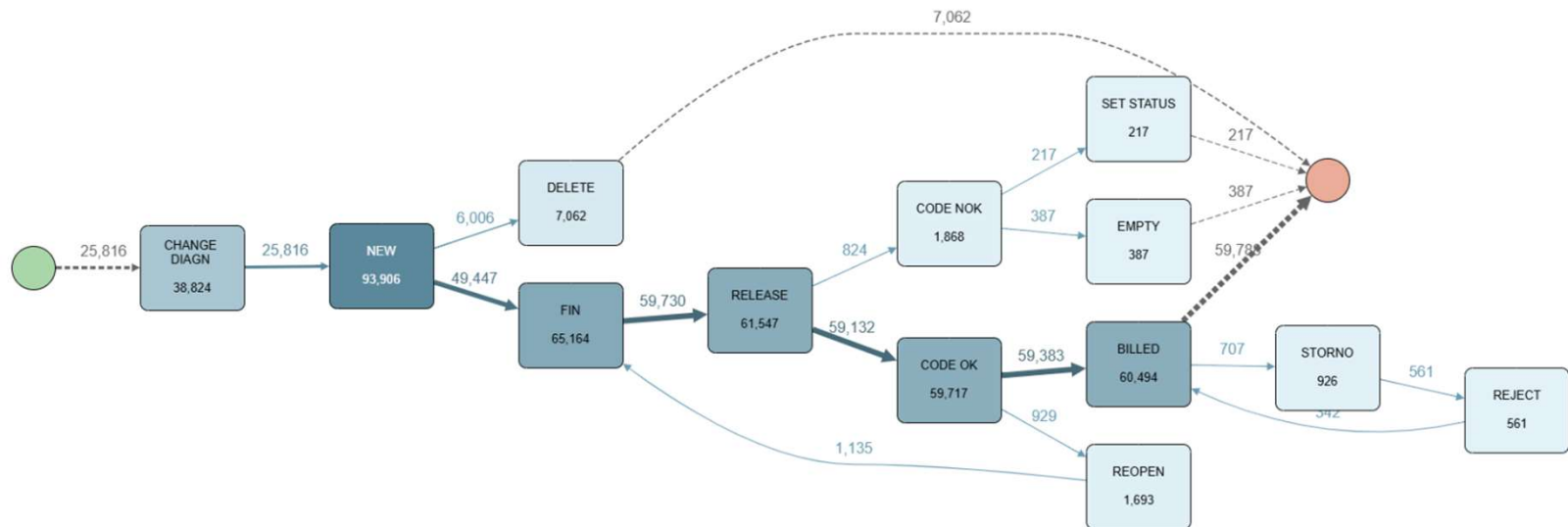
Project implementation process

- ✓ 1. Problem framing
- ✓ 2. Data collection

Meeting 2:

- 3. Process Analysis
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PROCESS OVERVIEW



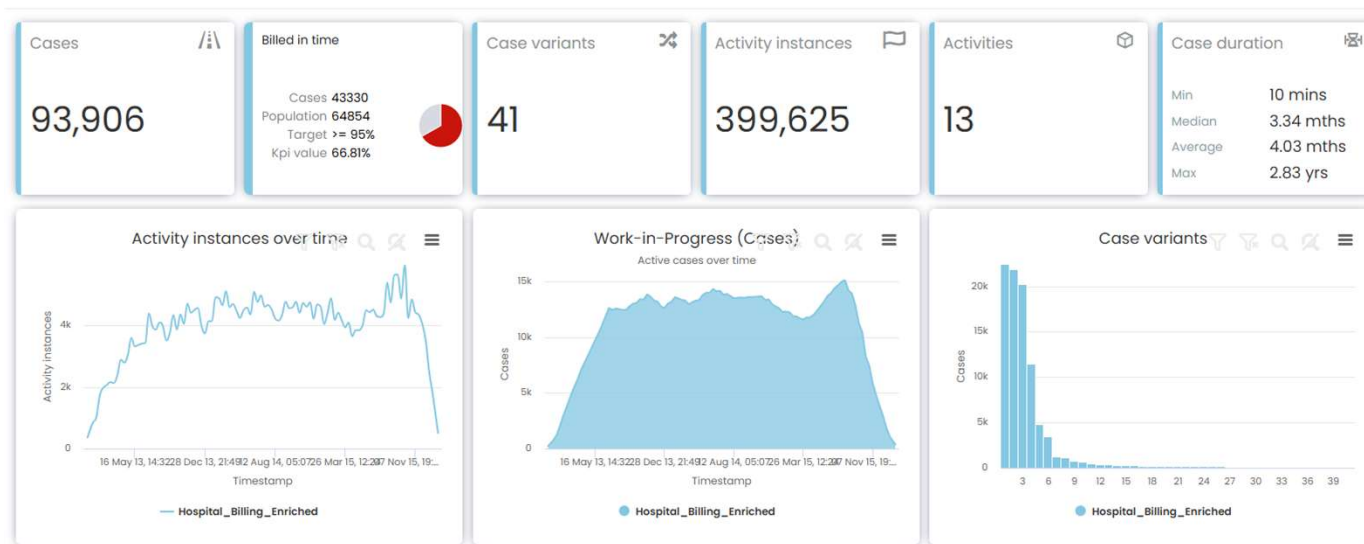
The Discovery Process map identified the primary billing pathway:

CHANGE DIAGN->NEW-> FIN -> RELEASE-> CODE OK-> BILLED

Reference path (focuses on operational flow):

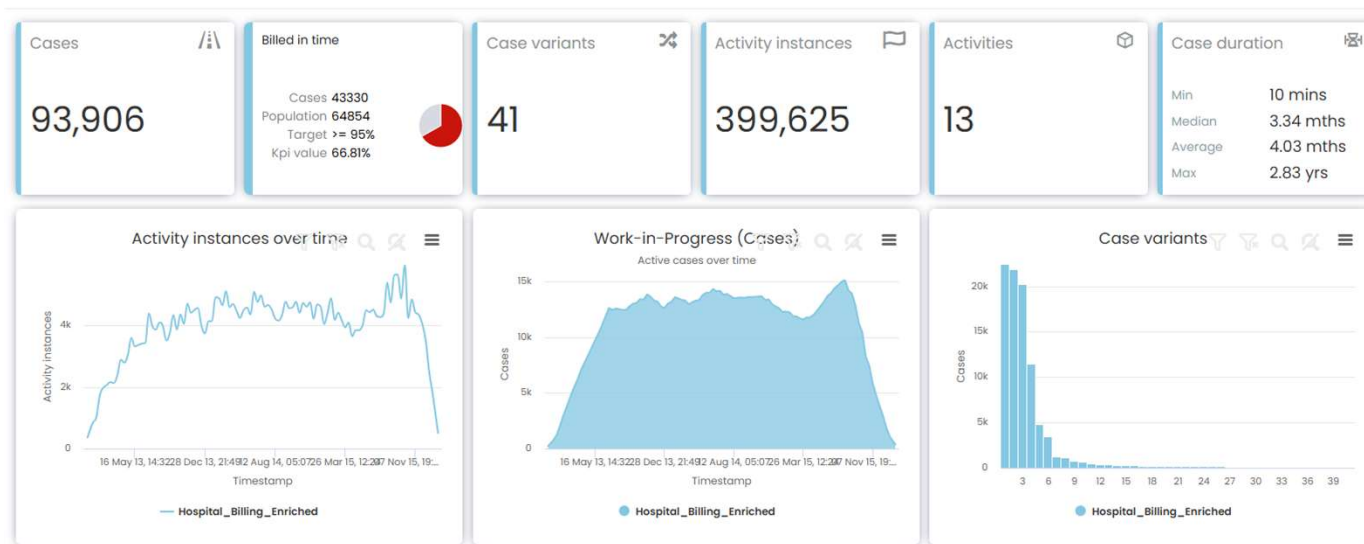
NEW-> FIN -> RELEASE-> CODE OK-> BILLED

DASHBOARD OVERVIEW



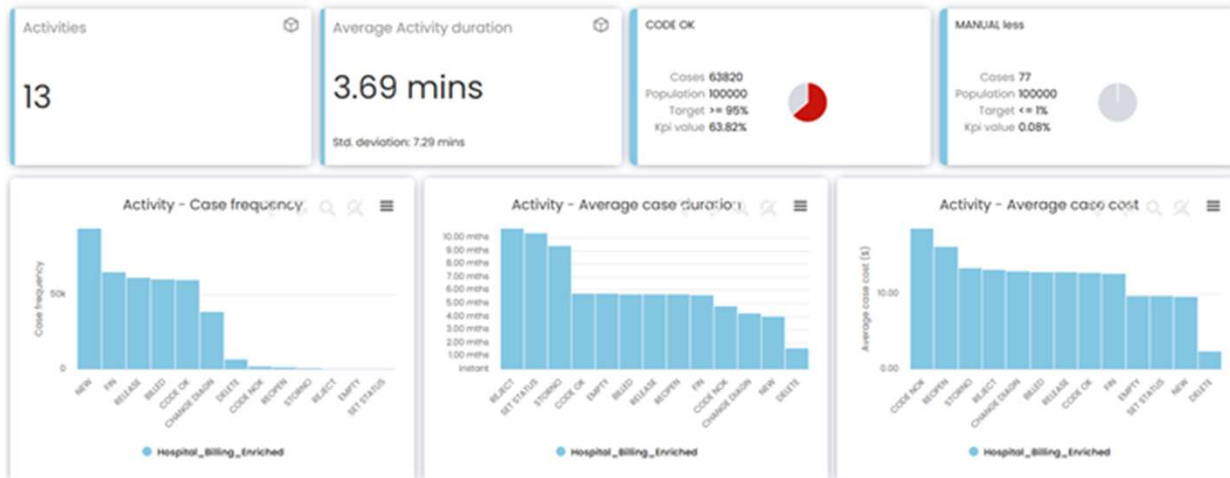
- 93.9K cases analysed; only **60.49k cases were billed**
- Average case duration **4.03 months** (max 2.83 yrs)
- Average case duration of complete cases is 5.74 months; the incomplete cases may influence the overall cycle time
- 41 variants → process inconsistency
- High delays in **CODE OK** (KPI 83.8%) and **FIN** activities
- Cost and duration trend together → rework drives cost

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DASHBOARD OVERVIEW



Activity-Level Insights

- 13 total activities analysed; **average duration = 3.7 mins**
- **CHANGE DIAGN** → most expensive rework (≈ **28% of total rework cost**)
- **RELEASE** also among top rework drivers
- **REOPEN KPI = 0.46%** (target ≤ 3%) → effective control on manual rework

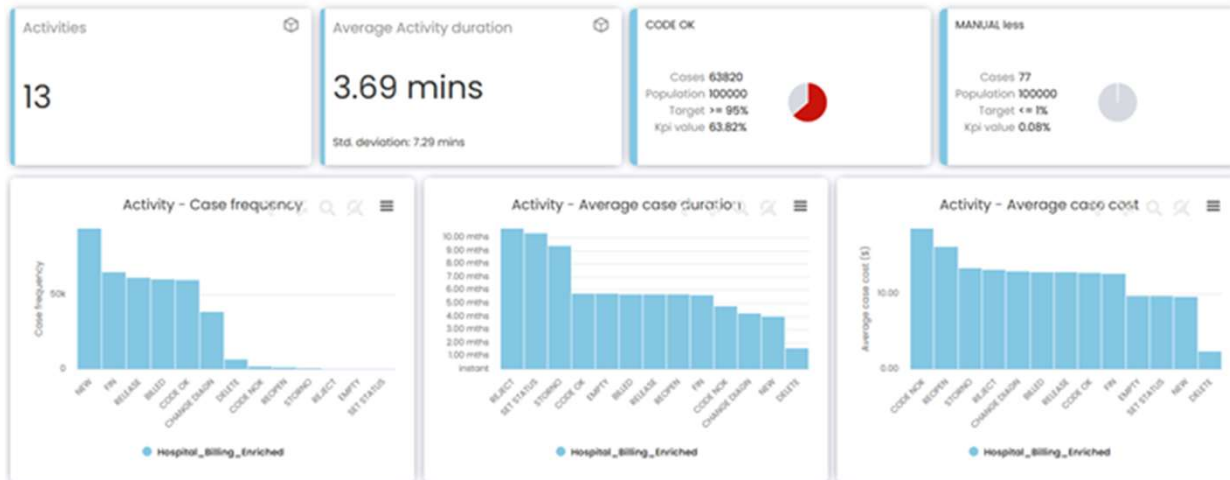
Resource-Level Insights

- 6 key roles: Admin, Medical Coder, Insurance Officer, Billing Specialist, Supervisor, System
- **Admin:** highest workload (**39% of cases**) & largest cost share (**27%**)
- **Insurance Officer:** longest average duration (**~10 months**) → external claim verification delays
- **System steps** add minimal time/cost → strong automation contribution

Interpretation

Activity and role analysis reveal that rework costs are concentrated in a few key transitions (CHANGE DIAGN & RELEASE), while human-driven delays (especially Insurance Officer tasks) remain the main performance constraint.

DASHBOARD OVERVIEW



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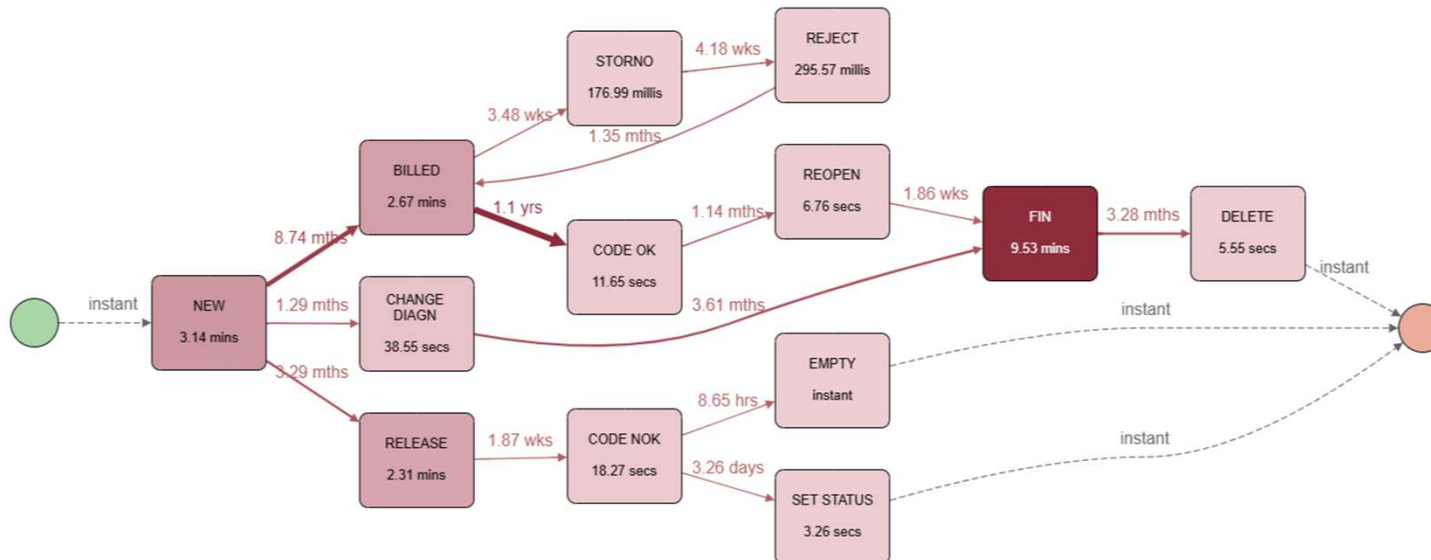
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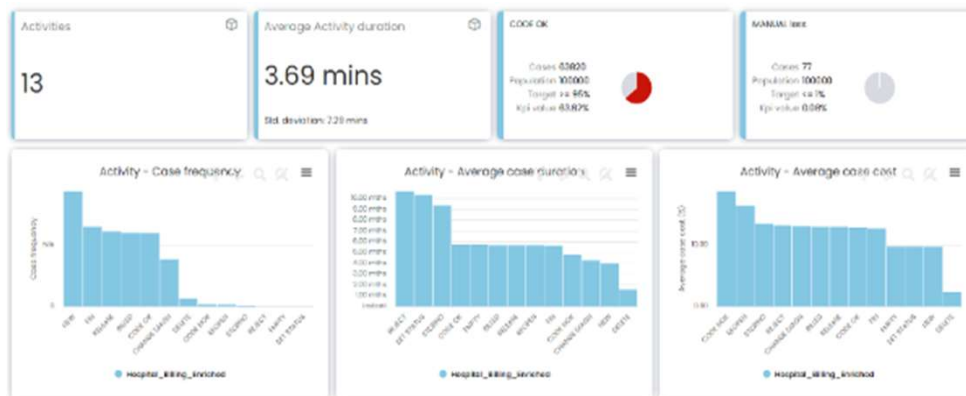
Bottlenecks



- A major waiting bottleneck occurs between **BILLED** -> **CODE OK**, adding **1.1 years** to some cases.
- The **FIN** activity (finalising the insUrance coverage) was identified as a key activity bottleneck, with a duration of **9.53 minutes** affecting throughout.
- Temporal conformance checking revealed 21,200 cases violating six months SLA with an average of 10.56 months indicating significant delays.

PAINPOINTS & PROBLEMS

Rework Loops



Rework was observed in several activities which substantially increased cost and time

- Billed rework: +5.54 months, + AUD 4.16 per case
- Fin rework: +2.83 months, + AUD 7.81 per case
- Release rework: +2.29 months, + AUD 8.04 per case
- CODE OK rework: + 5.48 months, + AUD 4.24 per case

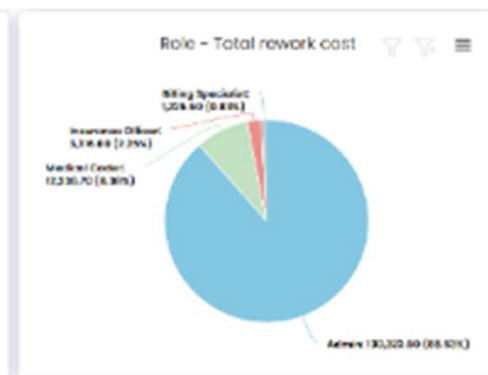
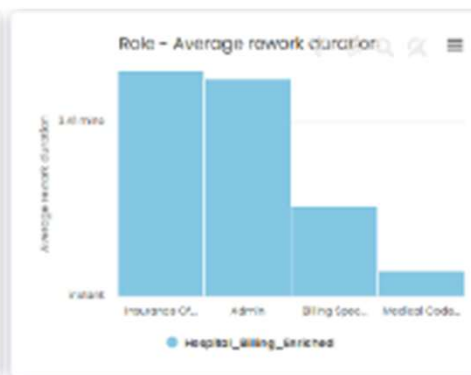
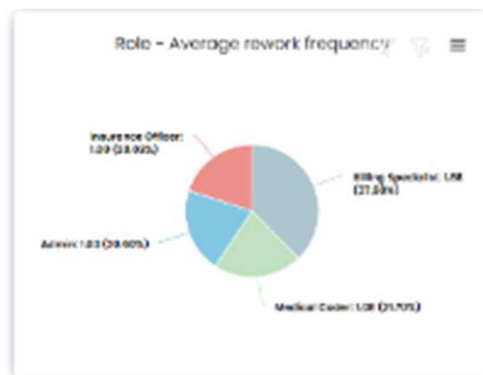
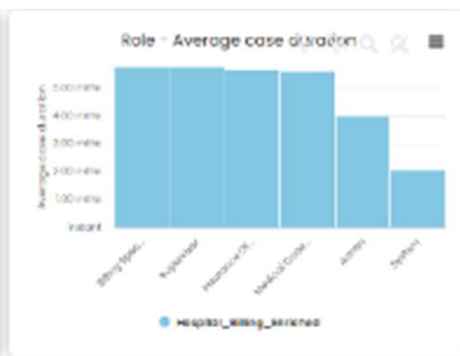
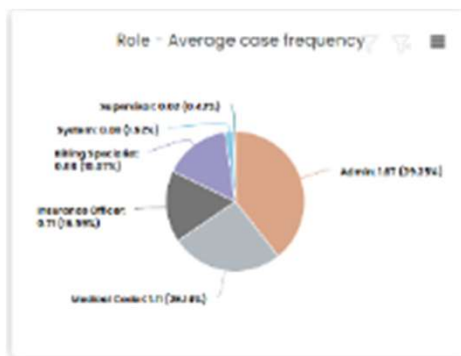
CHANGE DIAGN rework: +0.39 months, + AUD 4.01 per case

When all the rework is removed, duration drops to 3.87 months & cost is reduced by 9.44 AUD



PAINPOINTS & PROBLEMS

Cost Analysis



The **cost analysis** revealed:

Average activity cost: 9.66 AUD

Maximum activity cost: REOPEN (21.76 AUD)

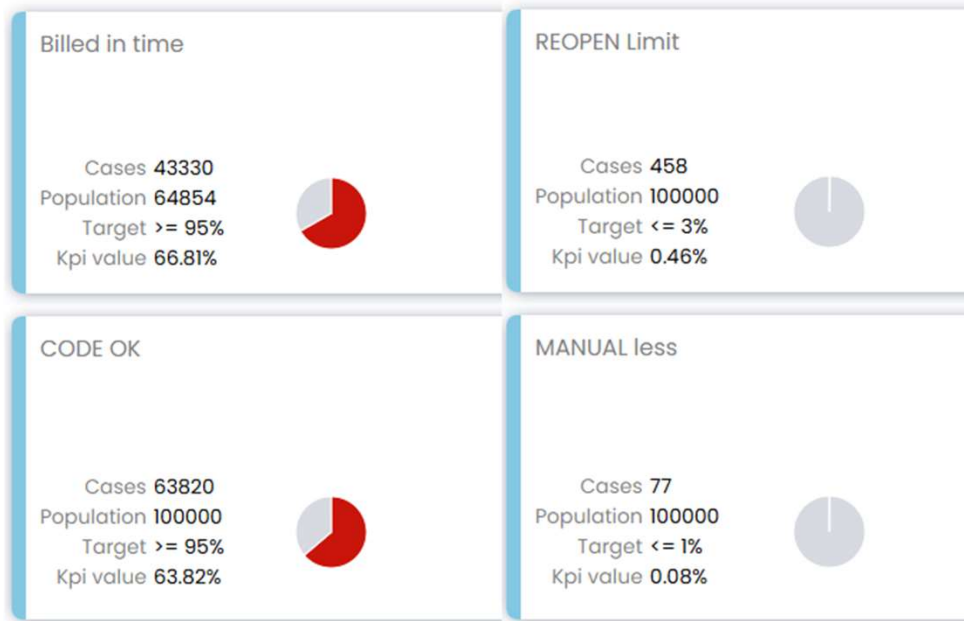
High role cost: Supervisor (21.67 AUD per hour)

Observation: The average cost of a case is 9.66 AUD, reflecting total expenditure greater than 900K AUD.

Explanation: Rework loops and Bottlenecks were the main drivers of cost increases with delays directly inflating administrative overhead

COMPLIANCE

KPI Metrics



SLA target: ≤ 6 months

- KPI target 95%
- Current KPI: 66.81%
- Failed
 - o Violations: 21.2K cases (avg 10.56 months)

Medical code is valid (CODE OK):

- KPI target $> 95\%$
- Current KPI: 63.82%
- Failed

Minimise REOPEN:

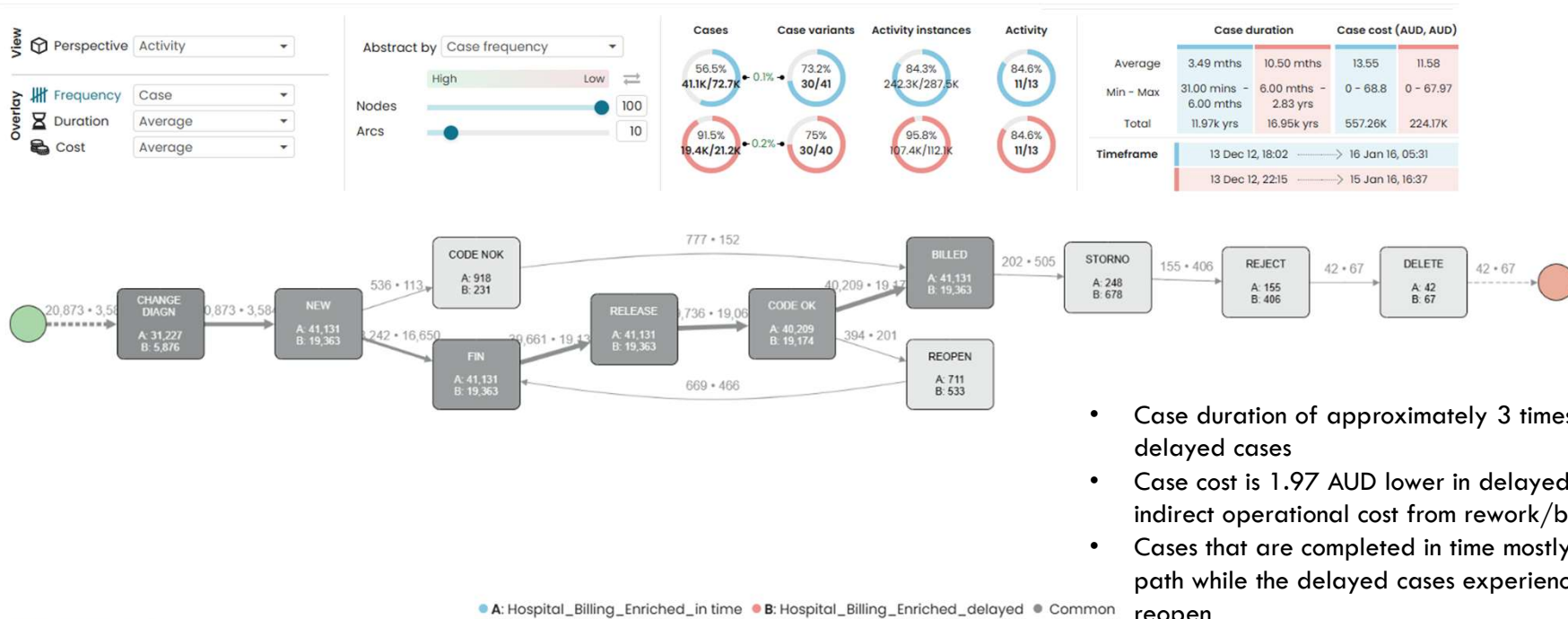
- KPI target $< 3\%$
- Current KPI: 0.46%
- Passed

Minimise Manual task:

- KPI target $< 1\%$
- Current KPI: 0.08%
- Passed

PAINPOINTS & PROBLEMS

Flow Comparison



- Case duration of approximately 3 times longer on average in delayed cases
- Case cost is 1.97 AUD lower in delayed cases (may include indirect operational cost from rework/bottleneck)
- Cases that are completed in time mostly follow the reference path while the delayed cases experienced more rework and reopen
- Delayed cases have higher relative percentage of activities that imply cancellation such as DELETE and REJECT

PAINPOINTS & PROBLEMS

Flow Comparison



- All roles take longer to process in cases which violated compliance
- Cases completed within 6 months suggest higher cost
- Total cost of on time cases are significantly higher as driven by larger number of cases

PAINPOINTS & PROBLEMS

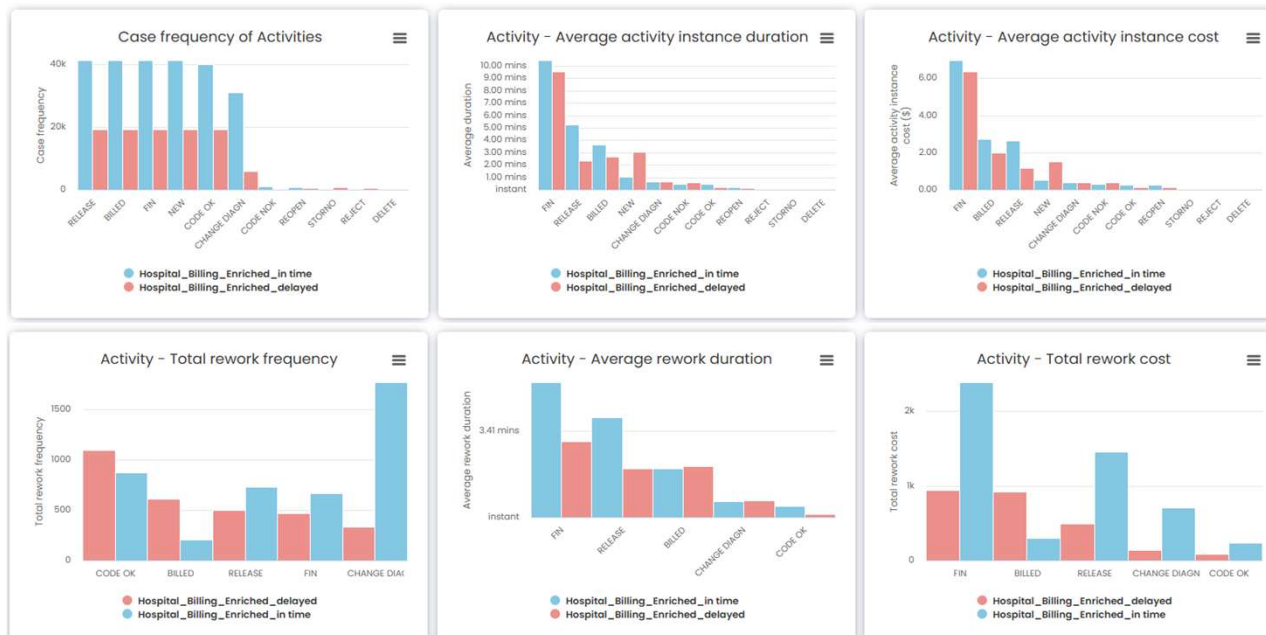
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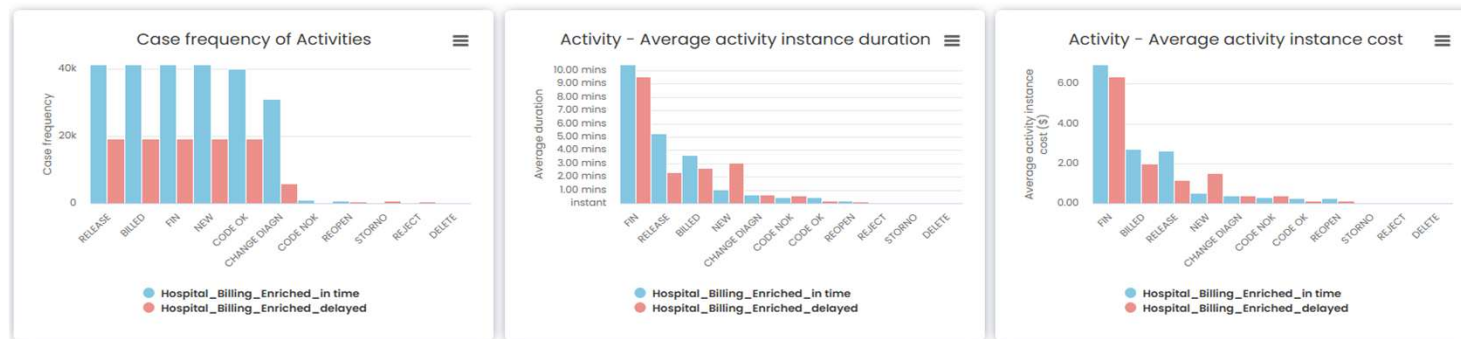
Flow Comparison



- Rework is detected more in CHANGE DIAGN activity despite taking less time overall, implying the activity does not have strong influence on the average cycle time
- Differences between total rework cost of each activity in two compared cases are remarkably high

PAINPOINTS & PROBLEMS

Flow Comparison



- Delays tend to occur in the FIN stage (Bottleneck) for both on time and delayed cases
- FIN cost and duration is the highest, suggesting its impact on both time and expense efficiency

FIN

- Costs over \$6 on average
- Takes approximately 9-10 mins and is significantly longer than other activities

PAINPOINTS & PROBLEMS

Flow Comparison

Case duration Hospital_Billing_Enriched_in time	Case duration Hospital_Billing_Enriched_in time
Min 5.46 days	Min 10 mins
Median 4.06 mths	Median 2.33 mths
Average 4.04 mths	Average 2.11 mths
Max 5.98 mths	Max 6 mths

Case duration Hospital_Billing_Enriched_delayed	Case duration Hospital_Billing_Enriched_delayed
Min 6.04 mths	Min 6 mths
Median 9.55 mths	Median 11.99 mths
Average 10.83 mths	Average 10.55 mths
Max 2.58 yrs	Max 2.83 yrs

- FIN are frequently reworked, especially in delayed cases indicating possible errors in coding which require correction
- CHANGE DIAGN rework is detected more among on-time cases despite taking less time overall, implying the activity does not have strong influence on the average cycle time.
- Rework at FIN is time-consuming and contribute to significant delays, while also costs the highest
- Without rework from FIN, the average case duration of delayed cases dropped by 0.28 months while dropped up to nearly 2 months in on-time cases

FINAL RECOMMENDATIONS



Automate FIN & Assignment of Medical Codes using AI to reduce time consumption and resource usage



Add SLA alerts in Apromore dashboard for cases > 6 months to improve SLA compliance while also set the solid rules of REOPEN cases



Enhance the efficiency of workload distribution by training supervisors to support admin and medical coders



Enhance the quality of initial validation by automating pre-validation at new application registration stage

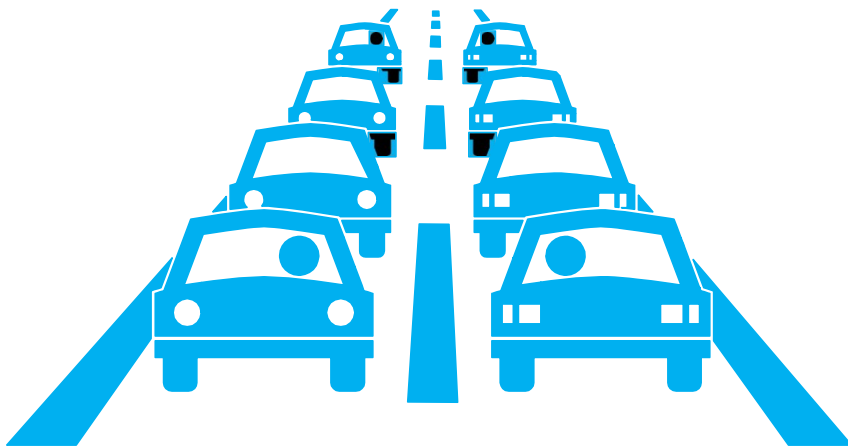
WHAT's NEXT



Next Steps for Hospital Billing Improvement

1. Approve automation pilot for FIN and Medical Coding modules (Q1 2026)
2. Staff training & workload balancing for Insurance and Supervisor roles
3. Deploy Apromore SLA dashboard for real-time monitoring
4. Review KPIs quarterly; aim for > 90 % SLA compliance by mid-2026

CONCLUSION



Insights gained

- Flow of activities
- Case durations
- Resource utilization
- Expenses

Delays caused by Manual

- Interventions Process
- Rework Resource
- Bottlenecks

What's next?

- Ongoing process monitoring and task automation are highly recommended for continuous improvement